

LOANS DATA ANALYSIS

Loans data across America for the 2015 year

INTRODUCTION

Welcome to our scientific poster, where we delve into the 2015 dataset of loans in America using a variety of approaches. Our study employs descriptive analysis, prediction, and clustering methods, each backed by meticulous data cleaning.

Throughout our exploration, we've generated insightful visual representations accompanied by clear explanations, shedding light on the intricacies of the data.

The poster not only showcases the outcomes of each method but also provides valuable observations, making it a comprehensive journey through the multifaceted landscape of loan data in the United States.



DATA

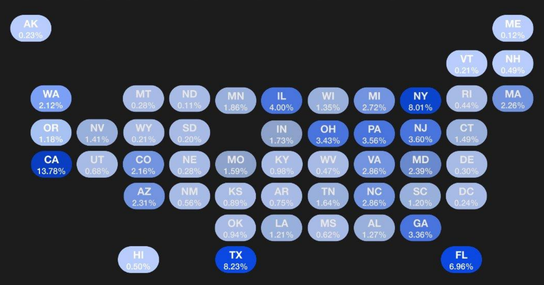
- **fico range low** — borrower's FICO boundary level
- **emp length** — availability of the loan in years
- **home ownership** — mortgage or rent or own
- **dti** — debt to income ratio
- **grade** — creditor's estimated grade rating for the loan
- **term** — duration of the loan (36 month, 60 month)
- **loan status** — status of the loan
- **funded amount** — loan amount
- **annual inc** — borrower's income
- **purpose** — purpose of the loan
- **employee title** — employee title of the borrower
- **state** — name of the state

METHODOLOGY

- **Data Filtering** — Removing non-meaningful data that is not utilized in the analysis, as it has the potential to adversely affect prediction accuracy.
- **Data Cleaning** — Utilized quantiles to eliminate outliers that could adversely affect prediction accuracy.
- **Descriptive Analysis** — Conduct a thorough analysis of the data using visualizations generated through PowerBI software, facilitating a comprehensive understanding of the dataset.
- **Standard Scaler** — Standard Scaler is used in machine learning to standardize the distribution, adjusting the distribution of each feature so that it has a mean of zero and a standard deviation of one.
- **Prediction** — Scaled data for consistency and used linear regression to establish relationships, improving accuracy in predictions.
- **Clusterization** — Clusterization was applied using the TF-IDF (Term Frequency-Inverse Document Frequency) Vectorizer to effectively organize and categorize employee titles.
- **Quantile filtering** — Quantile filtering has been used to avoid outliers by splitting the data into a basic percentage of data between 5% and 95% excluding data that is not in the outlier values.
- **Random Forest** — Random Forest is a popular algorithm that constructs multiple decision trees and outputs the class that is the mode of the classes of the individual trees, or the mean prediction for regression tasks.

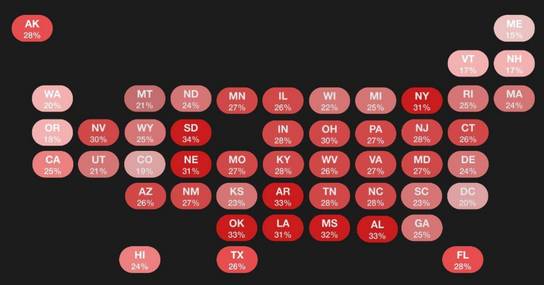
DESCRIPTIVE ANALYSIS

COUNT OF LOANS BY STATE



For 2015, California was the leader in the number of loans taken out, with Texas and New York in second and third place by a wide margin.

CHARGED OFF RATIO BY STATE



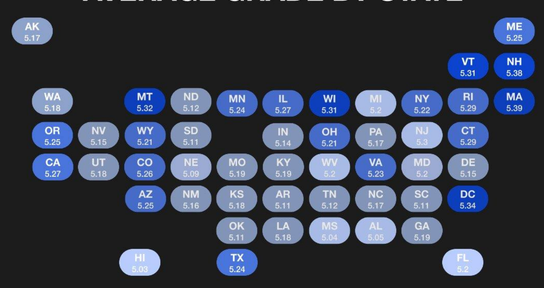
South Dakota, unfortunately, has the highest number of debtors who have defaulted on their loans. Also New York, which has one of the highest number of loans has a high failure rate.

DTI BY STATE

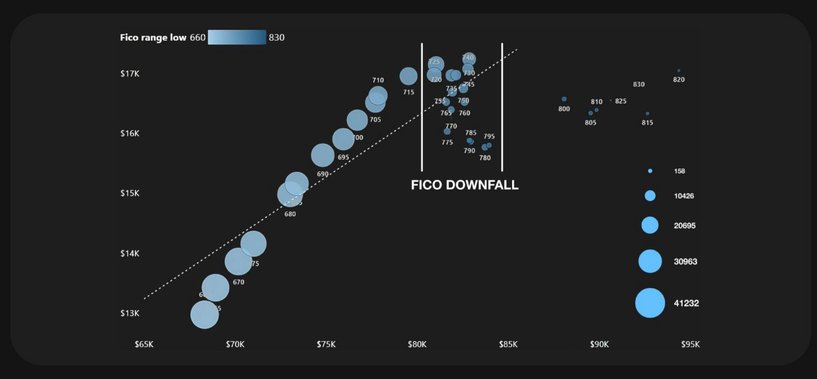


The DTI is fairly evenly distributed among the states, with South Dakota having the highest value, which correlates with its high level of charged coff.

AVERAGE GRADE BY STATE



The scores have little variation among states, ranging from 5 to 5.5 (C grade), indicating that lenders, on average, expect an approximately high probability of loan payment.



AVERAGE FICO BY INCOME AND LOAN

We can observe a steady increase in both wages and loan sizes as fico rises to 725, after which a sharp decline begins, which may be due to the fact that after fico becomes 730, lenders are satisfied with the size of earnings and focus on the size of the loan.

But after the fico passes the "dead zone" reaching very high values of 800 and above, the dispersion increases dramatically, which is true, as the loan size ceases to have any influence at all.

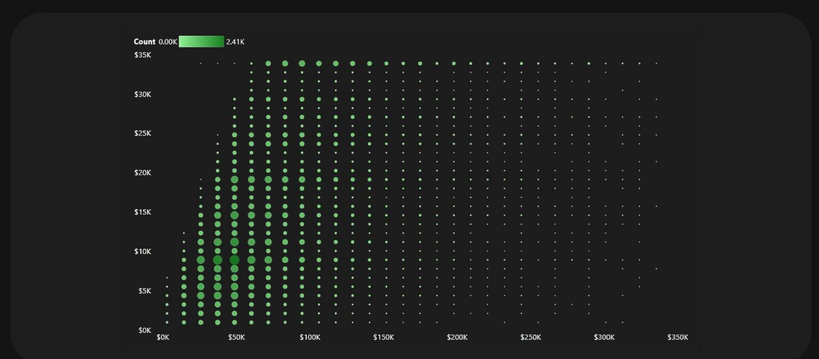
It can also be seen that as fico increases, the number of people with such a high indicator decreases.

FULLY PAID DEBT BY INCOME & LOAN

This chart demonstrates the distribution of the number of loans by the size of earnings and the loans themselves.

You can see from the graph that the dispersion is very well distributed, there is a specific edge after which no loans are taken at all.

Along with this, it is worth noting that the majority of loans are taken out by people with annual earnings of $50,000 to $8,000. This may be due to the fact that a large number of people have exactly this kind of earnings and need for emergency expenses.

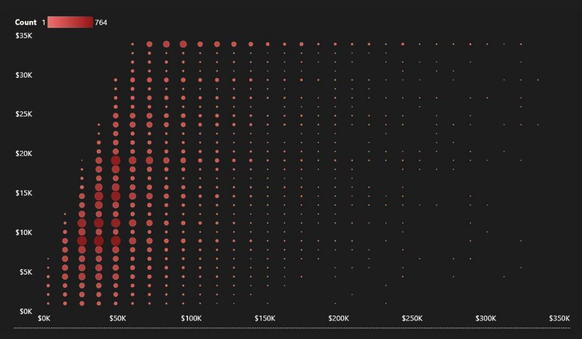


CHARGED-OFF DEBT BY INCOME & LOAN

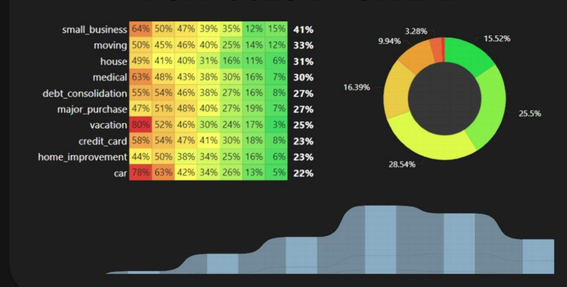
This graph, on the contrary, shows the number of people who have defaulted on the loan.

It can be seen that there are already far fewer such people, and one can see how, unlike the previous graph, the dots are flattened towards the right side, indicating that people are much more likely to default on a loan.

In addition, the bulk of people are shifted to the left relative to those who were able to pay off the loan in full. Finally, we have a detailed distribution of the number of loans for different combinations of loan size and earnings, but with common properties that have a large impact.

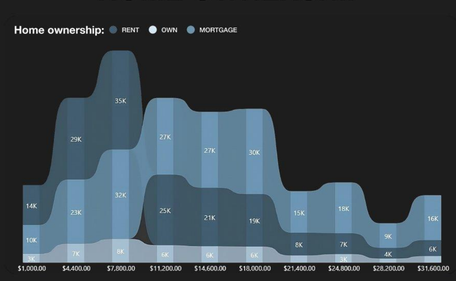


PURPOSES BY GRADE



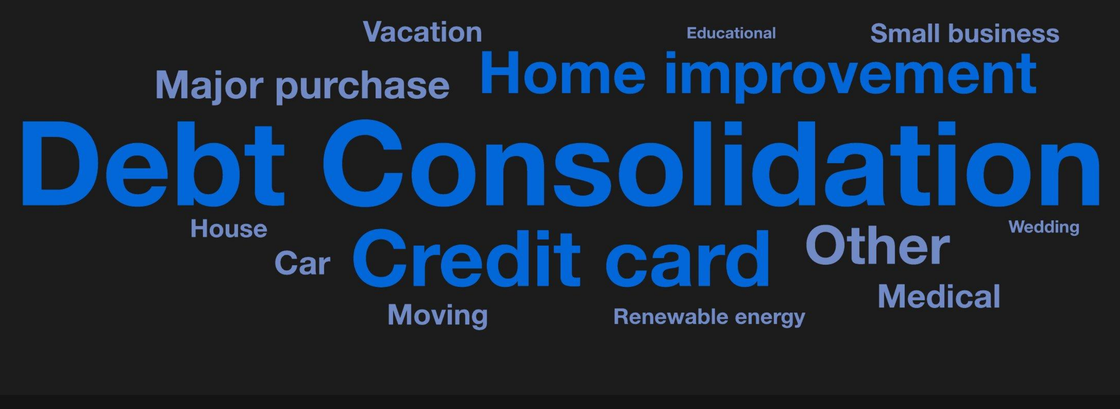
This combination of graphs explores the behavior of different purposes for loans and their chances of being paid with different scores. It can be seen that small business is the leader, while car and vacation loans with small appraisals are much more likely to fail. This is so because it is not only the percentage that has an impact, but also the sheer number of loans at a particular score. Thus we get negative skewness for the distribution of loans by grade.

HOME OWNERSHIP



In this graph, we see how people's loan amounts are distributed depending on home ownership. We can see that people renting houses have a lot more loans in the range of $1,000 to $11,000, after which mortgages dominate. Further, an important observation is that people who own housing generally rarely take out a loan, having a uniform distribution, so we can observe a low relationship with the size of the loan in this case.

The most common reason among them is debt consolidation, which accounts for 60% of all cases or approximately 25,000 applications. We assume that the reason for such an abundance of applications is the fact that debt consolidation allows a person who already has several loans with different terms and interest rates to combine them into one big loan.

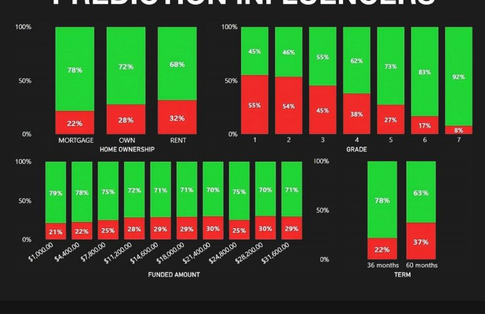


PREDICTION

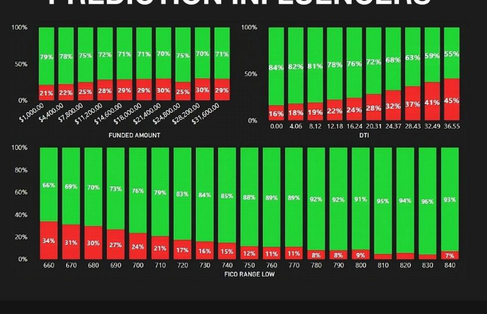
75.6%

Accuracy of Random Forest prediction

PREDICTION INFLUENCERS



PREDICTION INFLUENCERS



The prediction allowed us to obtain an accuracy of 75.6%. These graphs show which data influences our prediction more or less, from the estimates that have the greatest impact to different DTI levels and loan sizes. In the final we have fair values based on existing data. It is also important to note that we divided some data into dummy values, using a random forest to make predictions.

CLUSTERIZATION



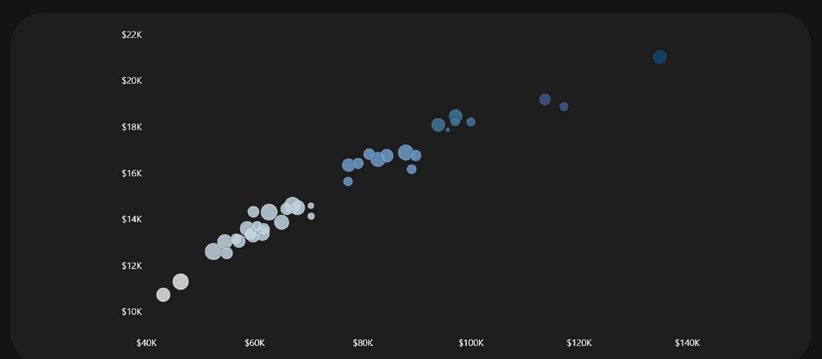
EMPLOYEE CLUSTER CLASSES

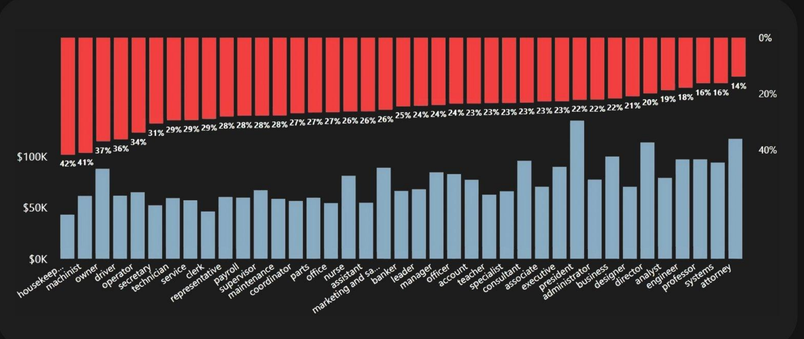
This graph shows us how, as the “prestige” of the position itself increases, its average earnings also grow, along with the size of the loan.

This linear movement allows us to see a high connection between the data. At the same time, we were also able to identify special job clusters among ready-made clusters.

One can see how different positions are divided into "castes". At the same time, the size of the dots also suggests that prestige is not affected by the number of different positions.

As a result, we have a very important observation of different positions.





EMPLOYEE INCOME AND CHARGED-OFF DISTRIBUTION

The graph shows how the earnings of people with different positions grow and what impact the position itself has on the rate of failed loans.

We can see a positive relationship between these values, which tells us that the position of the person taking out the loan has real weight on his success.

A striking example is that housewives are much more likely to lose credit than lawyers.

Thus, we obtain an observation on the basis of which further conclusions can be drawn.